Computational Omics in Gut Tumors and Beyond

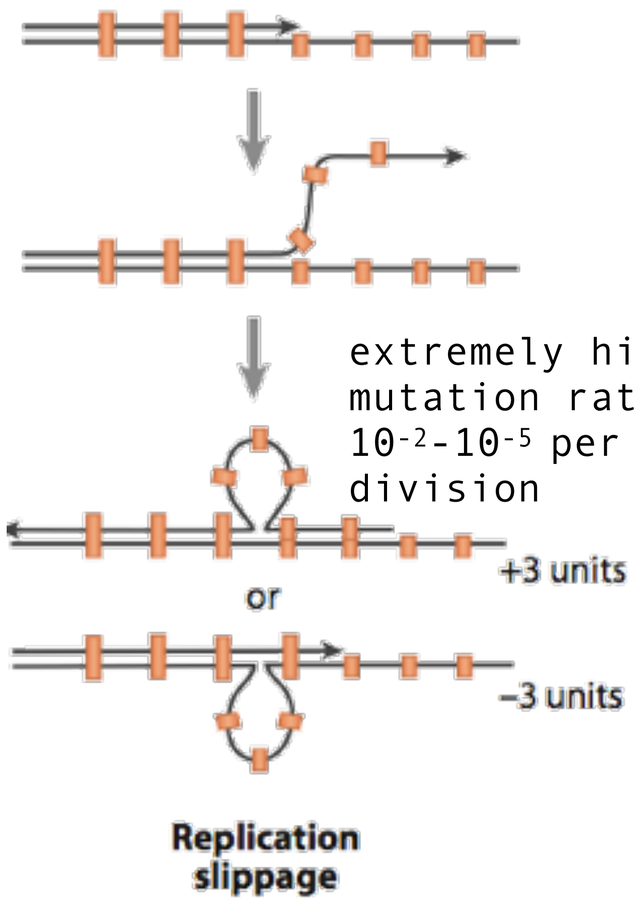
Research Centre for Bioinformatics

Tugce Bilgin Sonay, ZHAW

DNA Tandem Repeats

DNA Tandem Repeat Mutations

Also known as microsatellites



extremely high mutation rates: 10^{-2}-10^{-5} per cell division

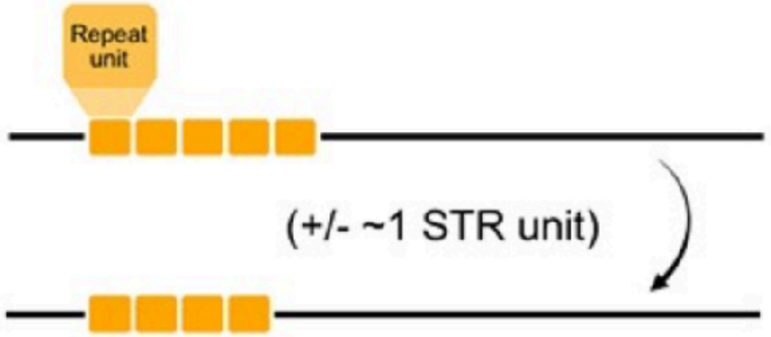
2 trillion cell divisions in your body every day

20 million to 20 billion mutations in your body every day

Gamayel *et al., Annu. Rev. Genet.*, 2010

DNA Tandem Repeats

Stepwise mutations

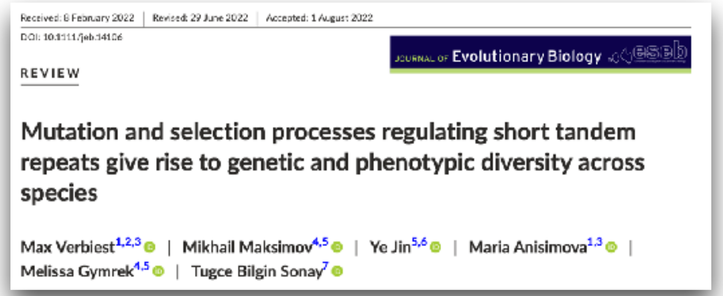


Very common, mostly benign, contribute to complex traits

Pathogenic repeat expansions



Rare, highly deleterious, (e.g. Huntington's, Fragile X)


Received: 8 February 2022 | Revised: 29 June 2022 | Accepted: 1 August 2022
DOI: 10.1111/jeb.14106
JOURNAL OF Evolutionary Biology
REVIEW

Mutation and selection processes regulating short tandem repeats give rise to genetic and phenotypic diversity across species

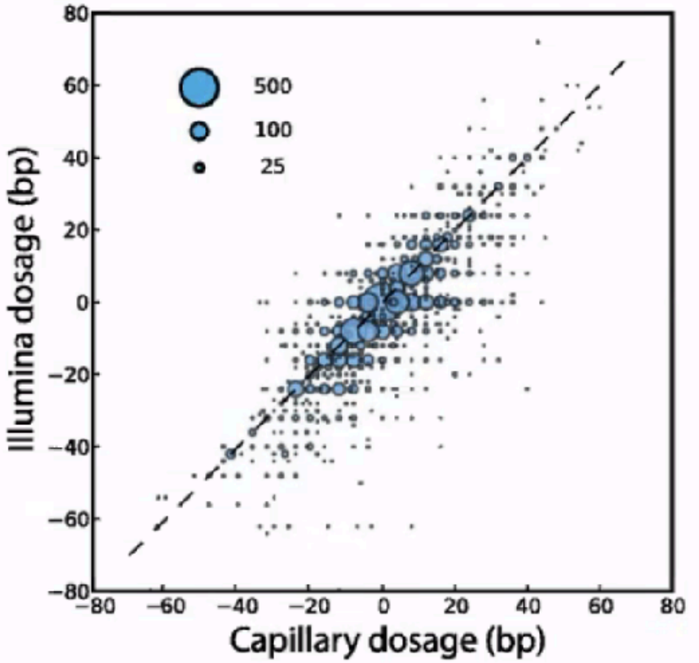
Max Verbiest[1,2,3] | Mikhail Maksimov[4,5] | Ye Jin[5,6] | Maria Anisimova[1,3] | Melissa Gymrek[4,5] | Tugce Bilgin Sonay[7]

Verbiest et al. Bilgin Sonay, JEB, 2022

STR calling has gotten better!

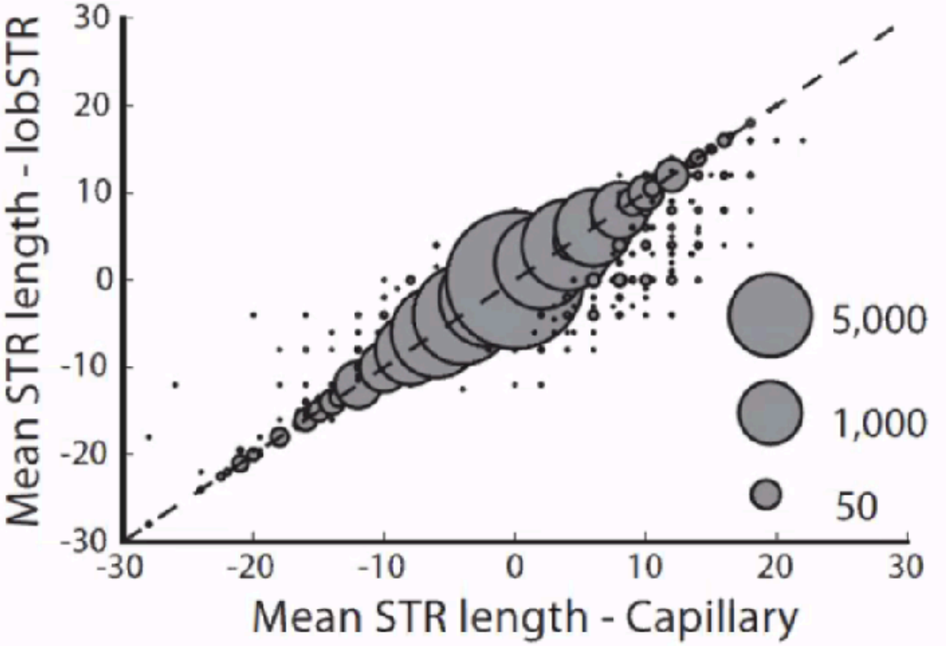
1000 Genomes (4x, lobSTR) (2014)

41% concordance with capillary data



SGDP (30x, PCR free, lobSTR) (2016)

93% concordance with capillary data

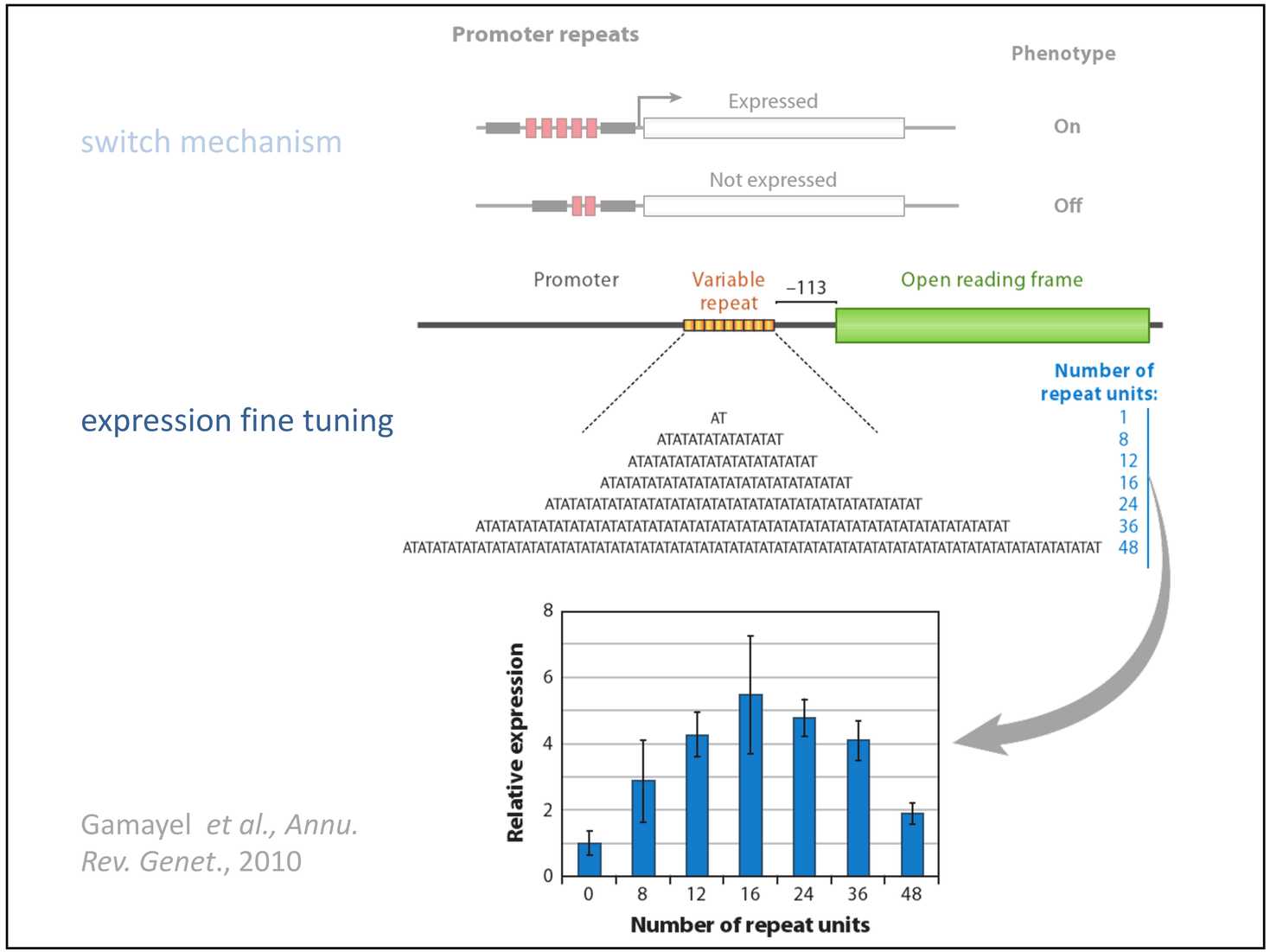


SGDP (30x, PCR free, **HipSTR**): 98.5% concordance on duplicated samples (2017)

DNA Tandem Repeats

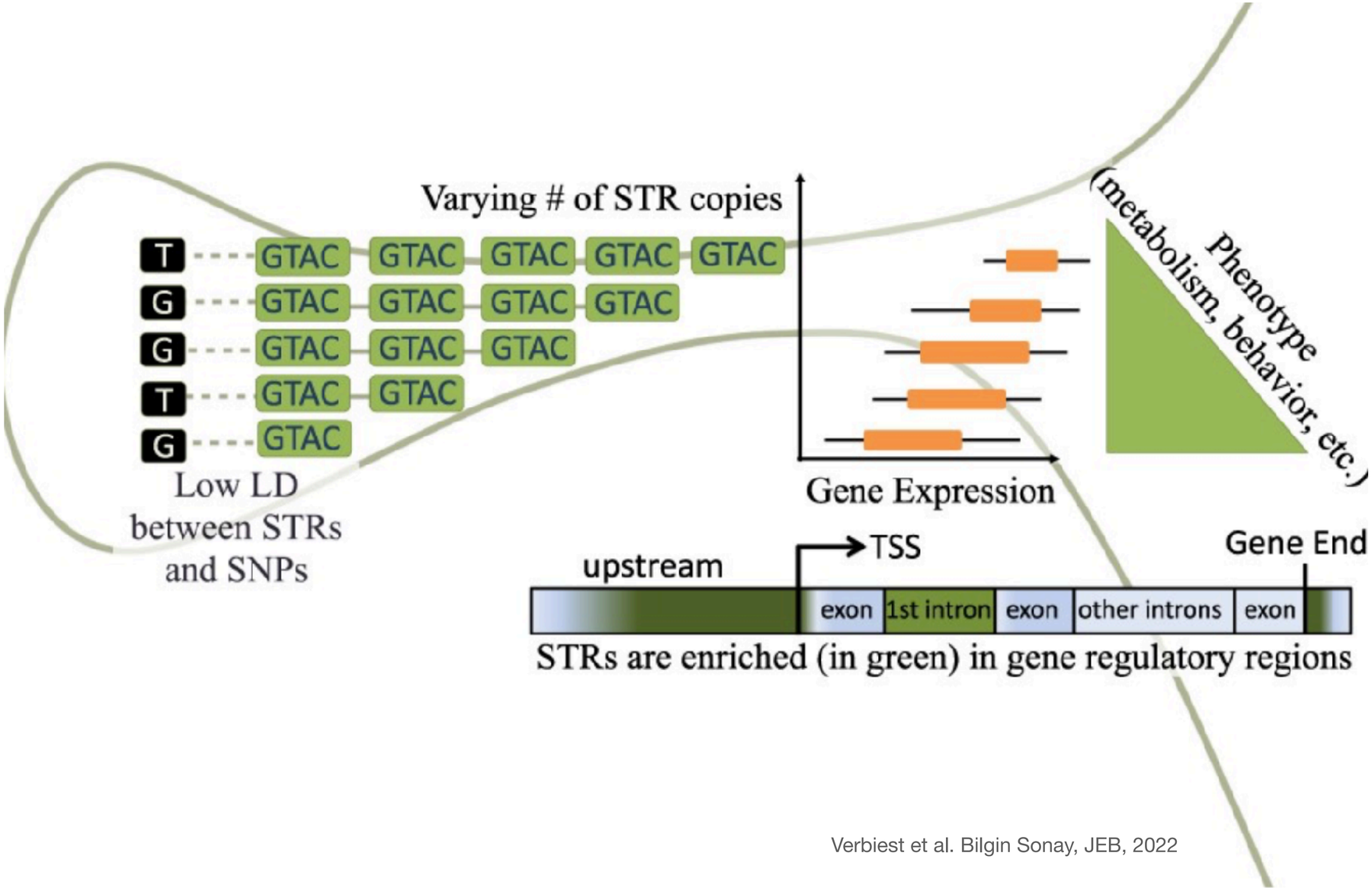
switch mechanism

expression fine tuning



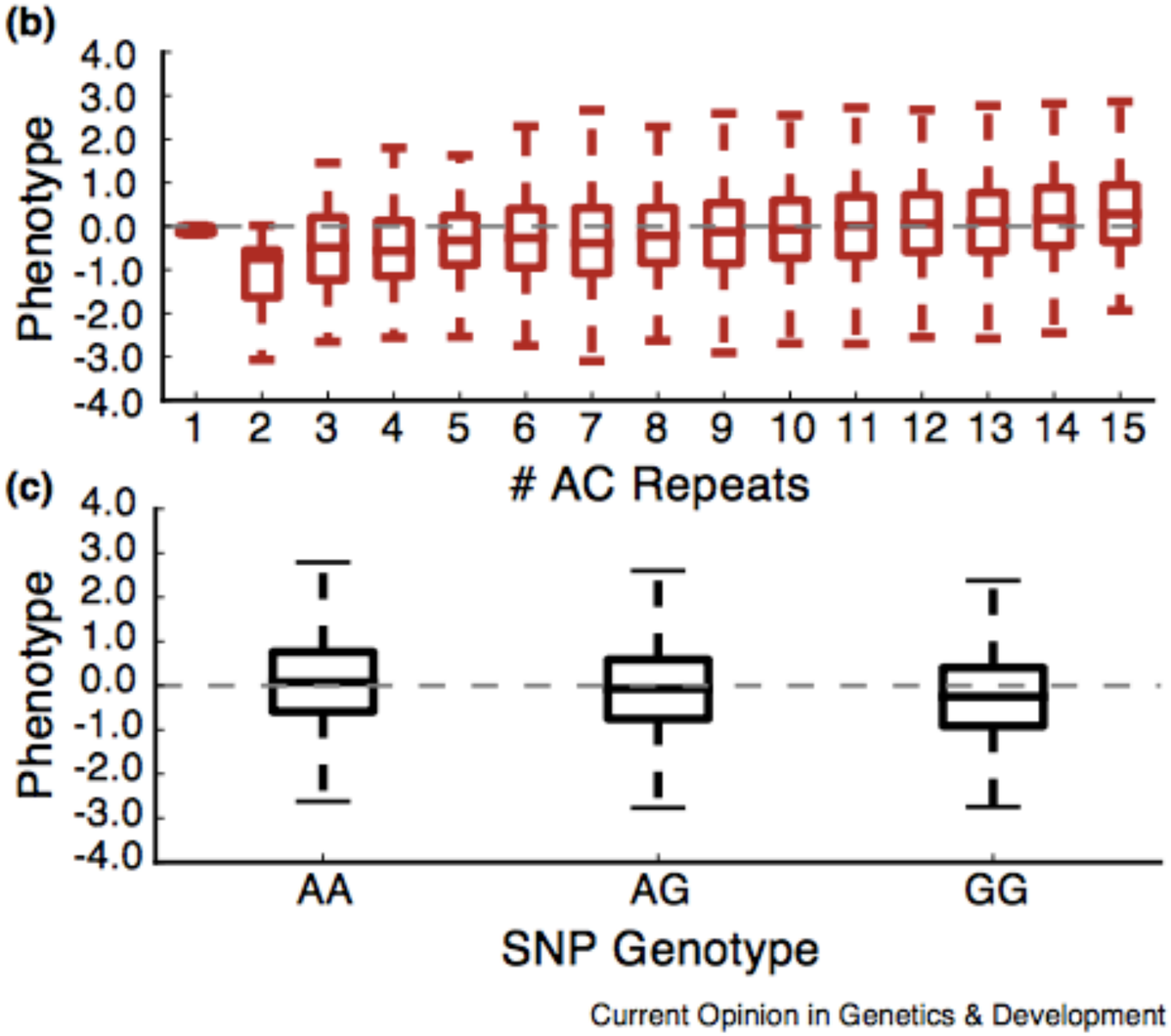
Gamayel *et al., Annu. Rev. Genet.*, 2010

DNA Tandem Repeats



Verbiest et al. Bilgin Sonay, JEB, 2022

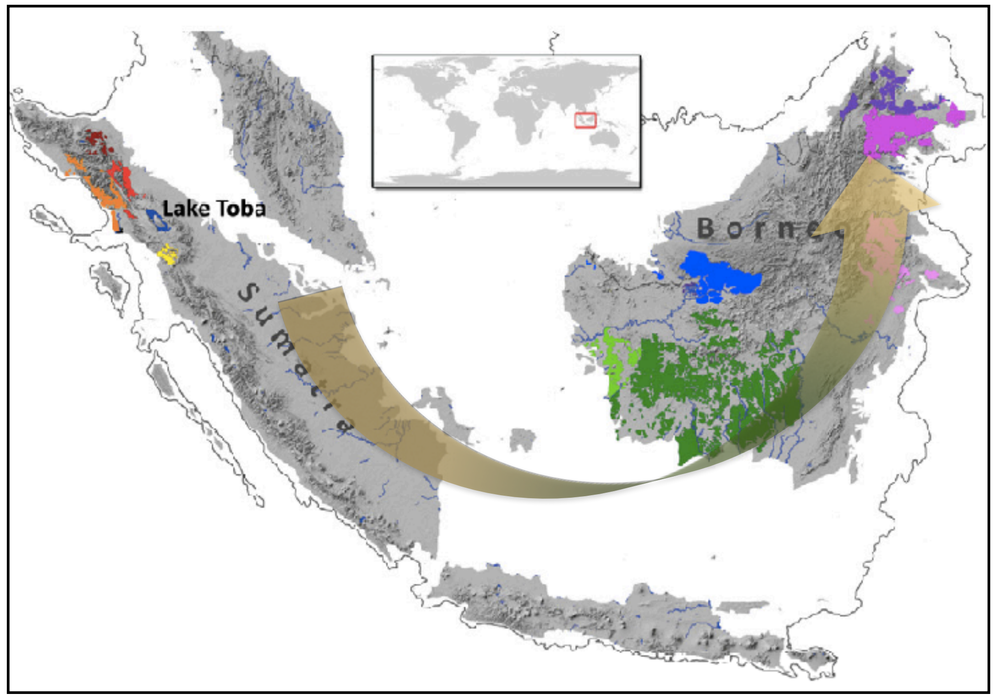
Short Tandem Repeats offer much more genotypic variation than SNPs



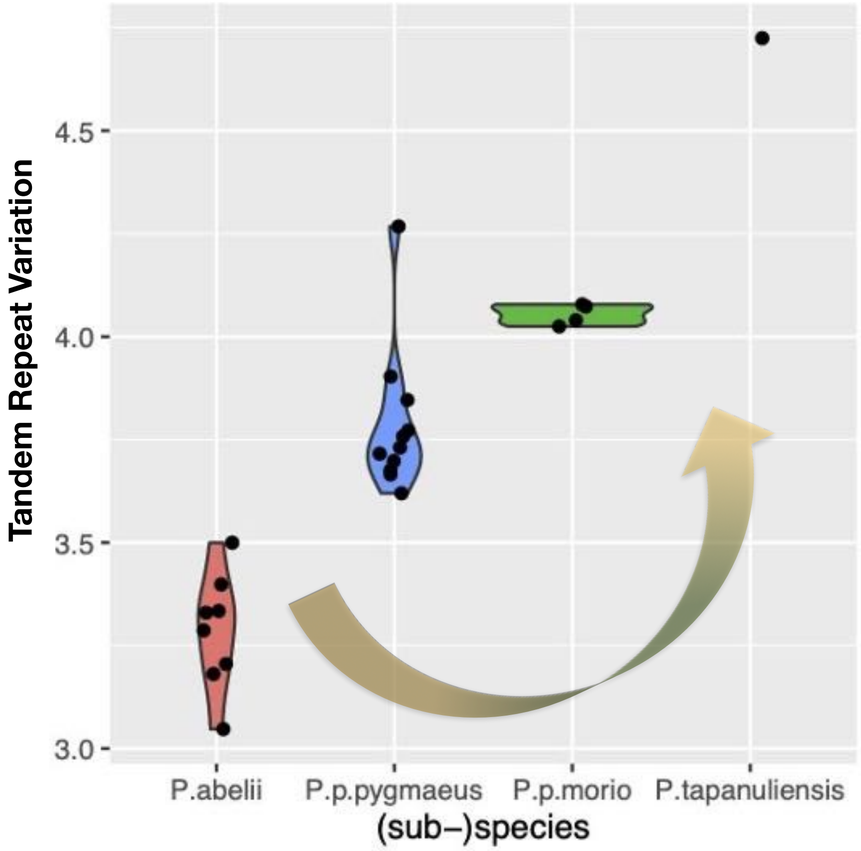
Current Opinion in Genetics & Development

Tandem Repeats capture recent evolution





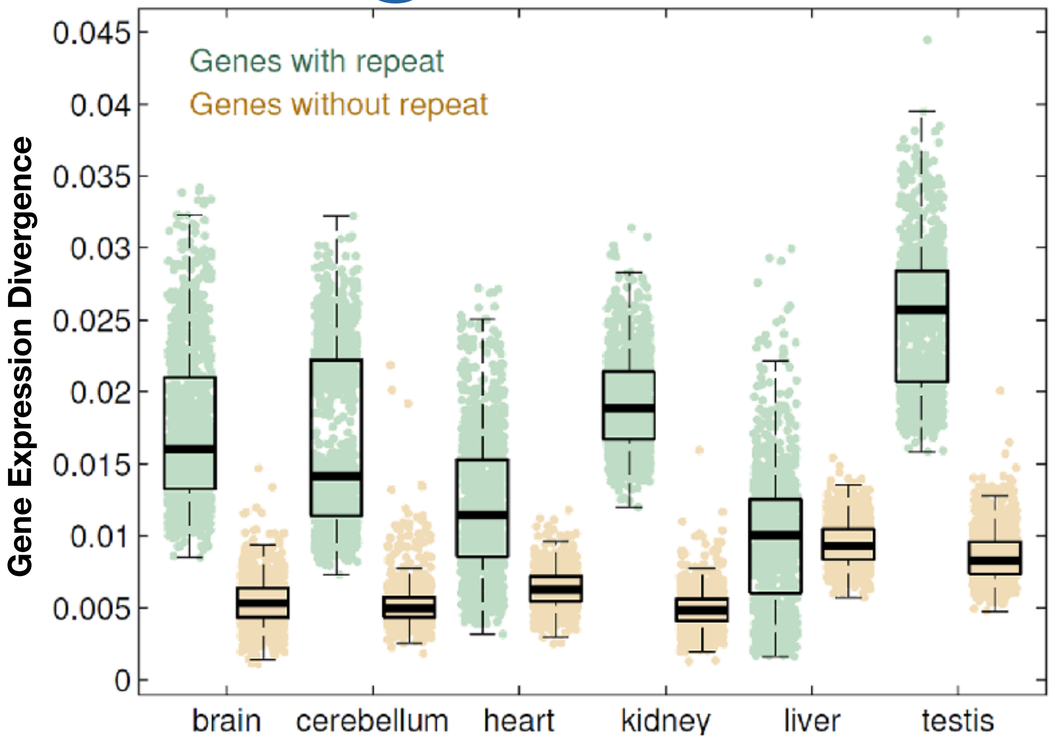
UZH, Anthropology Department



Evolutionary Distance from the origin of species

Voicu, Krutzen, Bilgin Sonay, 2022

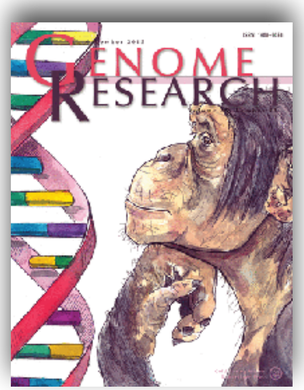
Tandem Repeats create phenotypic divergence

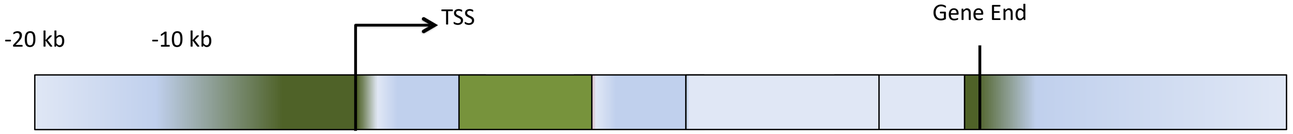


Research

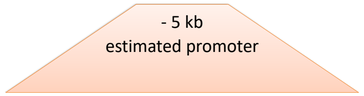
Tandem repeat variation in human and great ape populations and its impact on gene expression divergence

Tugce Bilgin Sonay,[1,2,11] Tiago Carvalho,[3,11] Mark D. Robinson,[2,4] Maja P. Greminger,[5] Michael Krützen,[5] David Comas,[3] Gareth Highnam,[6] David Mittelman,[6] Andrew Sharp,[7] Tomàs Marques-Bonet,[3,8,9,11] and Andreas Wagner[1,2,10,11]





Tandem repeats found in green regions associate significantly with higher gene expression divergence.

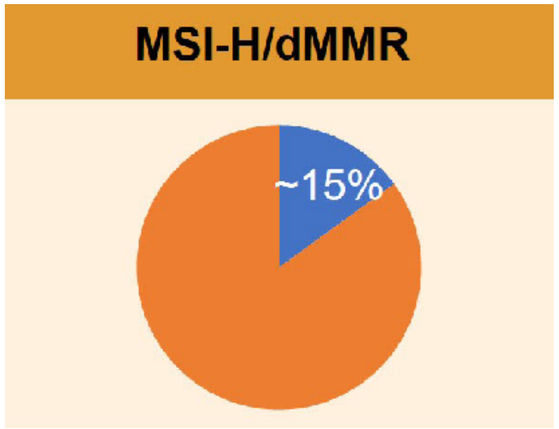


Tandem Repeats mutations are abundant in some cancers



diagnostic for a subtype of colorectal cancer also known as MSI-H MicroSatellite Instability High

The molecular mismatch repair system is defect -> repeat mutations



Colorectal Cancer (CRC)

- 3rd most common cancer worldwide
- 2nd leading cause of cancer-related deaths worldwide
- Nearly half of all new cases now occurring in those under 65
- In Switzerland, more than 4,000 people are diagnosed with colorectal cancer every year.
- Intestinal polyps and chronic inflammatory bowel diseases such as ulcerative colitis or Crohn’s disease can greatly influence the risk.

Some History

1990's: A certain set of tumors typically do not respond well to the standard fluorouracil (5-FU) chemotherapy.

Letter | Published: 10 June 1993

Ubiquitous somatic mutations in simple repeated sequences reveal a new mechanism for colonic carcinogenesis

Yurij Ionov, Miguel A. Peinado, Sergei Malkhosyan, Darryl Shibata & Manuel Perucho

Nature **363**, 558–561 (1993) | Cite this article

Abstract

SPONTANEOUS errors in DNA replication have been suggested to play a significant role in neoplastic transformation and to explain the chromosomal alterations seen in cancer cells[1]. A defective replication factor could increase the mutation rate in clonal variants arising during tumour progression, but despite intensive efforts, increases in tumour cell mutation rates have not been unambiguously shown[2]. Here we use an unbiased genomic fingerprinting technique[3] to show that 12 per cent of colorectal carcinomas carry somatic deletions in poly(dA . dT) sequences and other simple repeats. We estimate that cells from these tumours can carry more than 100,000 such mutations. Only tumours with affected poly(dA . dT) sequences carry mutations in the other simple repeats examined, and such mutations can be found in all neoplastic regions of multiple tumours from the same patient, including adenomas. Tumours with these mutations show distinctive genotypic and phenotypic features. We conclude that these mutations reflect a previously undescribed form of carcinogenesis in the colon (predisposition to which may be inherited) mediated by a mutation in a DNA replication factor resulting in reduced fidelity for replication or repair (a 'mutator mutation').

Some History

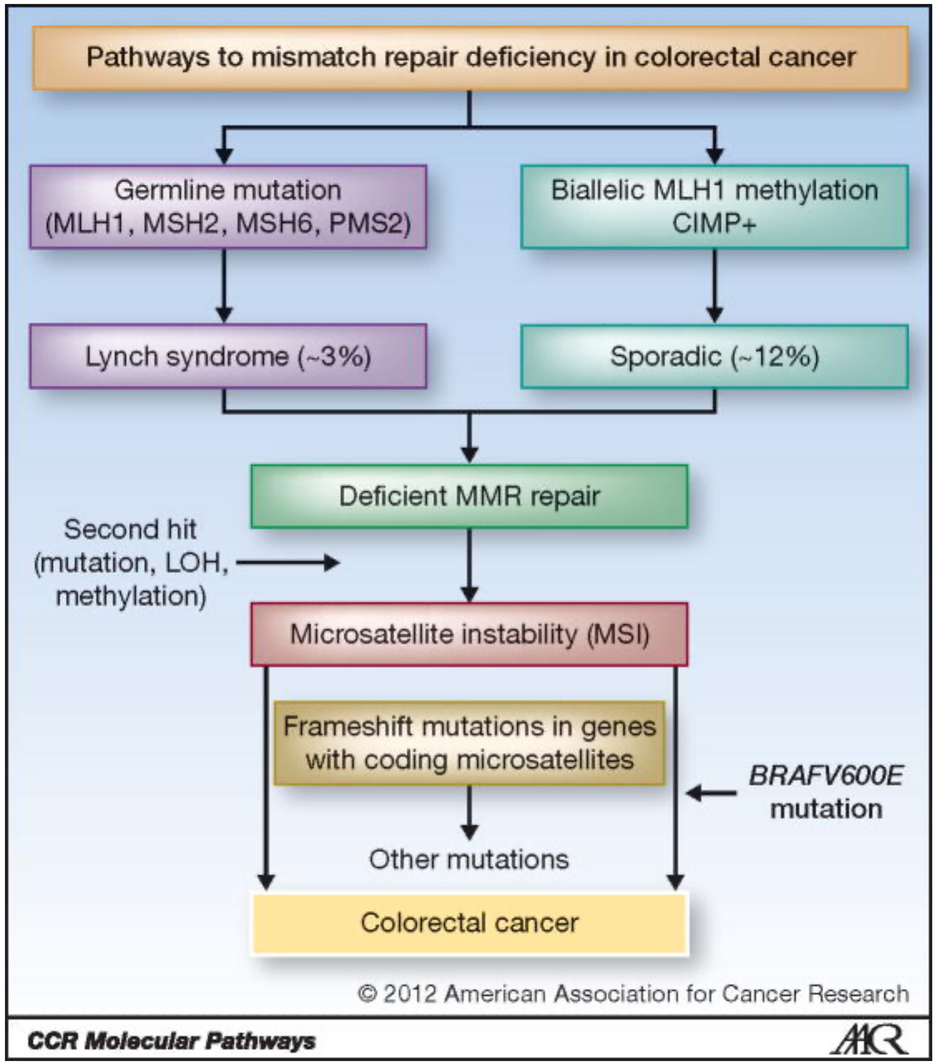
2000's: Microsatellite Instability testing becomes a vital tool

Microsatellite instability (MSI) is the molecular fingerprint of the deficient mismatch repair (MMR) system that characterizes approximately 15% of CRCs. MSI-High (H) tumors have better stage-adjusted prognosis but do not respond to

Published in final edited form as: Clin Cancer Res. 2012 Feb 2;18(6):1506–1512. doi: 10.1158/1078-0432.CCR-11-1469

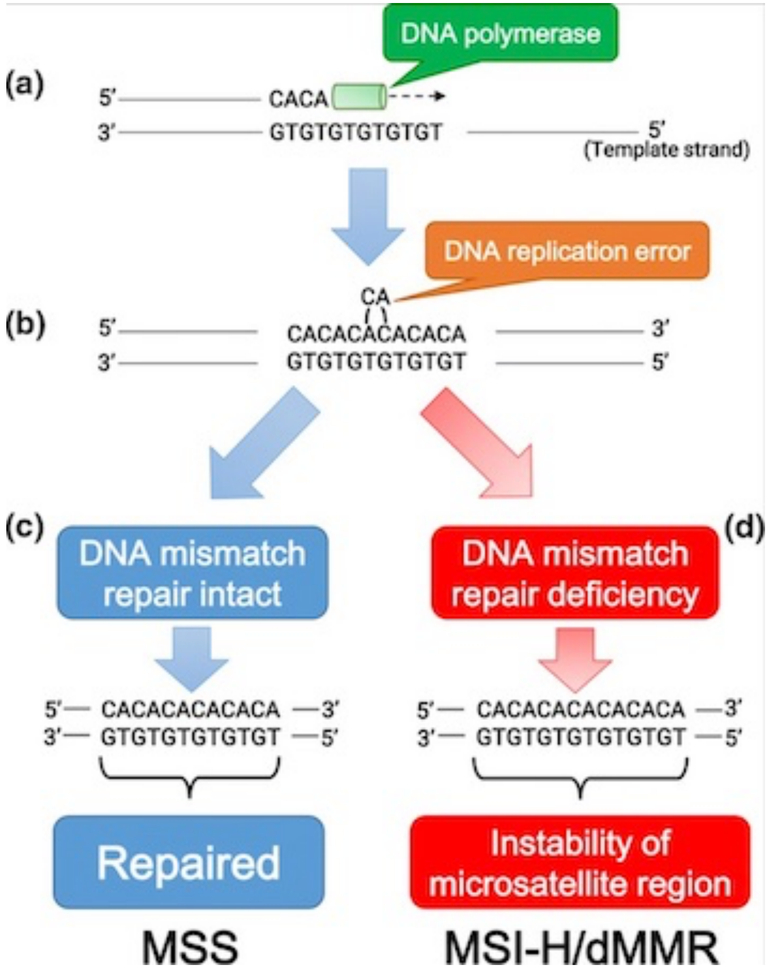
Molecular Pathways: Microsatellite Instability in Colorectal Cancer: Prognostic, Predictive and Therapeutic Implications

Frank A Sinicrope [1], Daniel J Sargent [2]



Some History

2000's: Without functioning MMR proteins, errors accumulate rapidly, especially in microsatellites (Short Tandem Repeats, STRs). Tumors with this defect are classified as **MSI-High (MSI-H)**



Some History

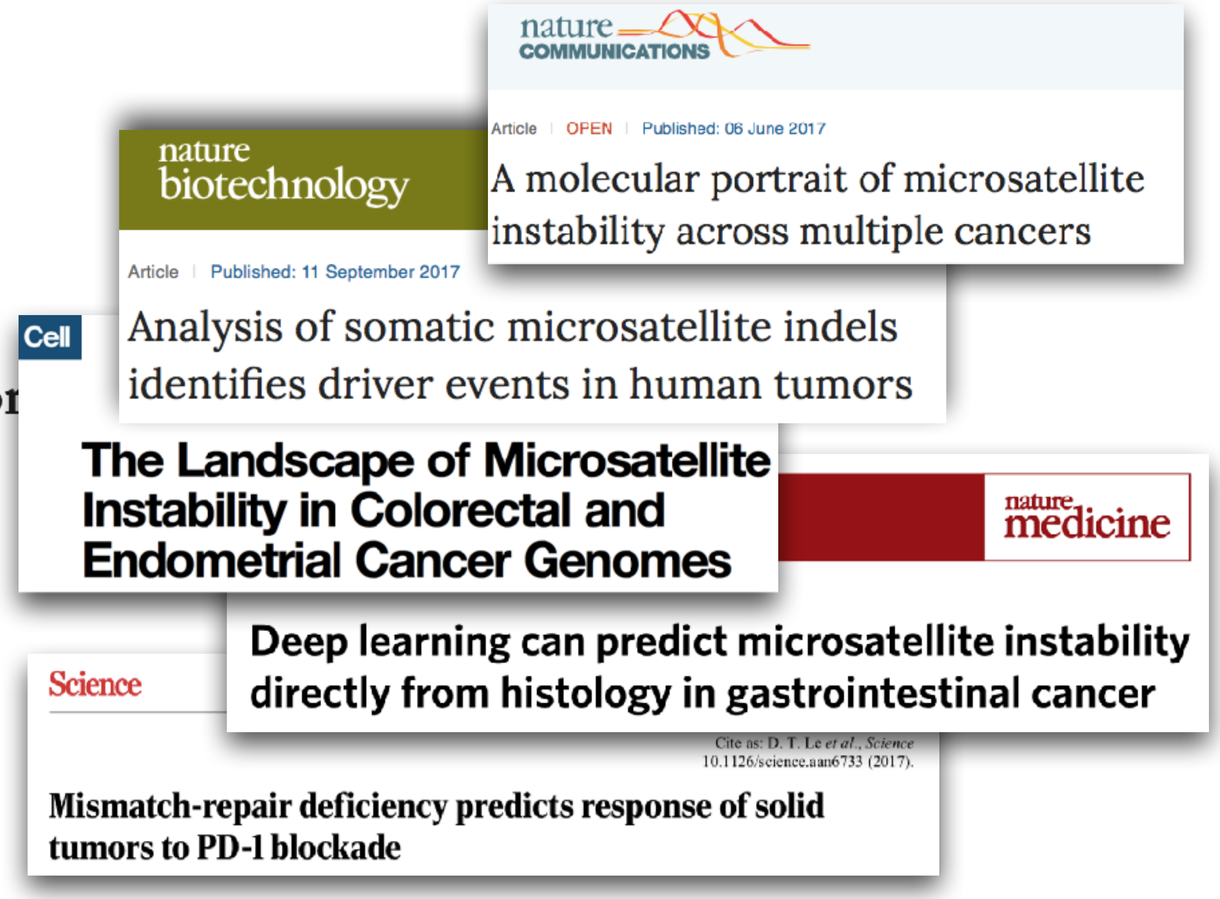
2010's:

genomic surveys

BMC Genomics. 2015 Sep 16;16(1):702. doi: 10.1186/s12864-015-1902-9.

A survey of tandem repeat instabilities and associated gene expression changes in 35 color cancers

Tugce Bilgin Sonay [1 2], Malamati Koletou [3], Andreas Wagner [4 5 6]

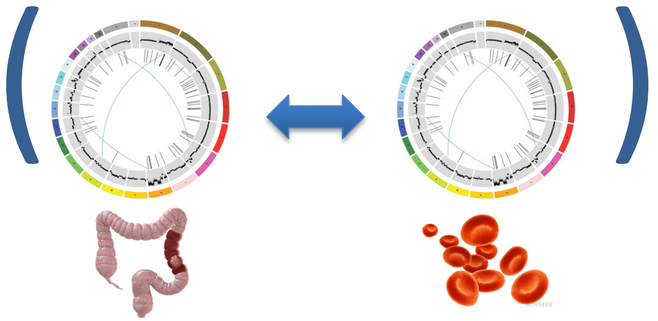


Tandem Repeats mutations are abundant in some cancers

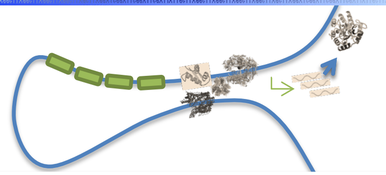


Data Set:

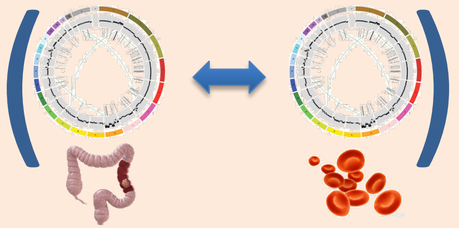
THE CANCER GENOME ATLAS



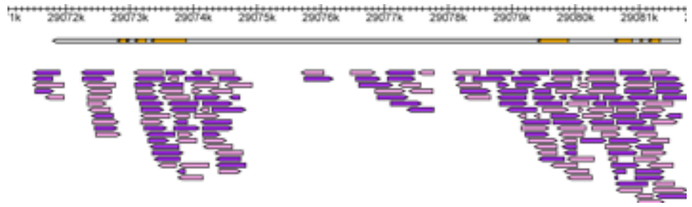
TANDEM REPEATS FINDER



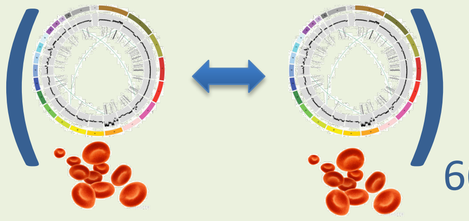
Tumor/normal genome pairs



RNA-seq for 19,000 genes in 37 tumors



Normal/normal genome pairs



Bilgin Sonay et al., BMC Genomics, 2015

Our first attempts to link to immunotherapy



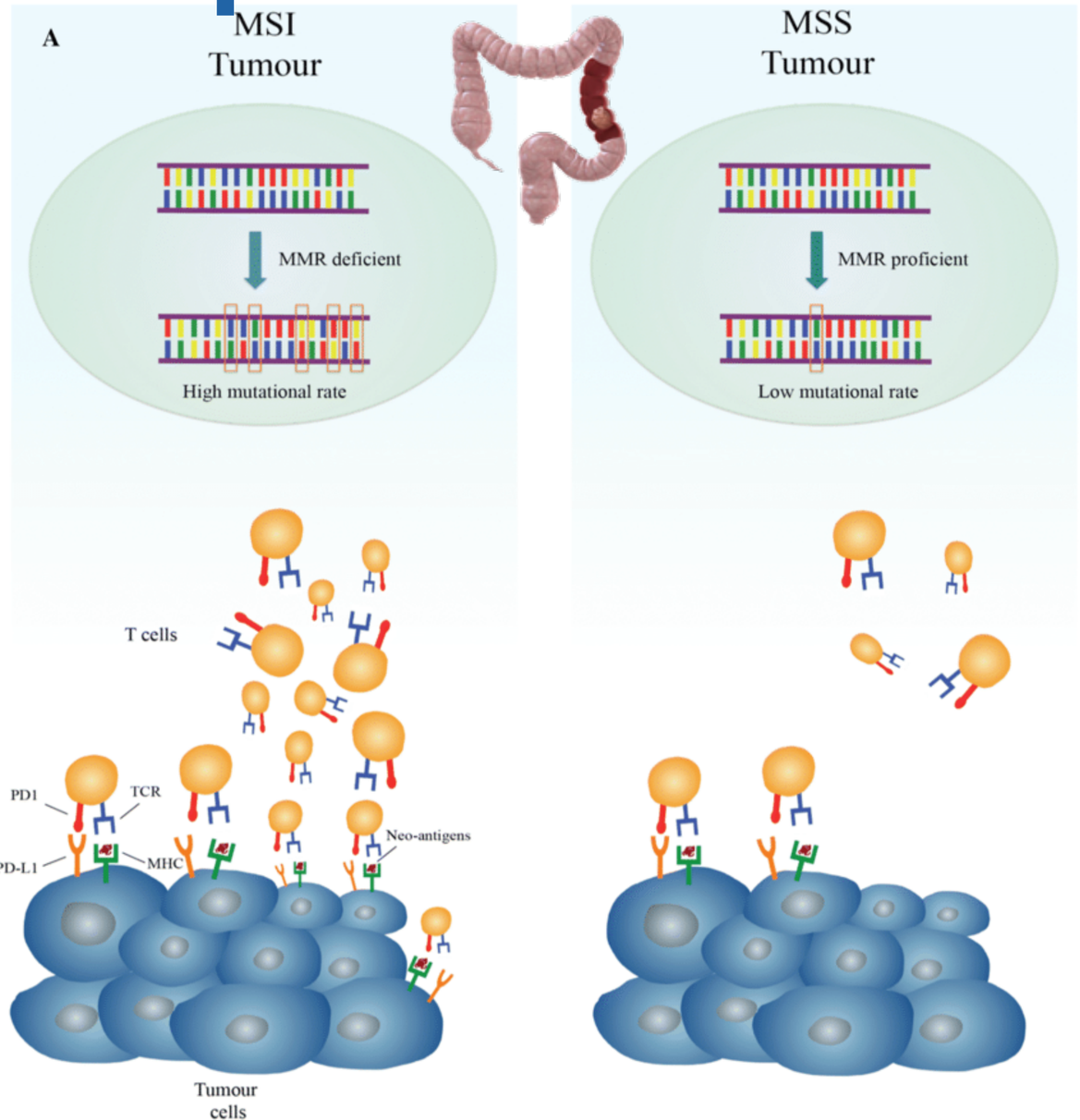
Maria Anisimova



Maria Rodriguez



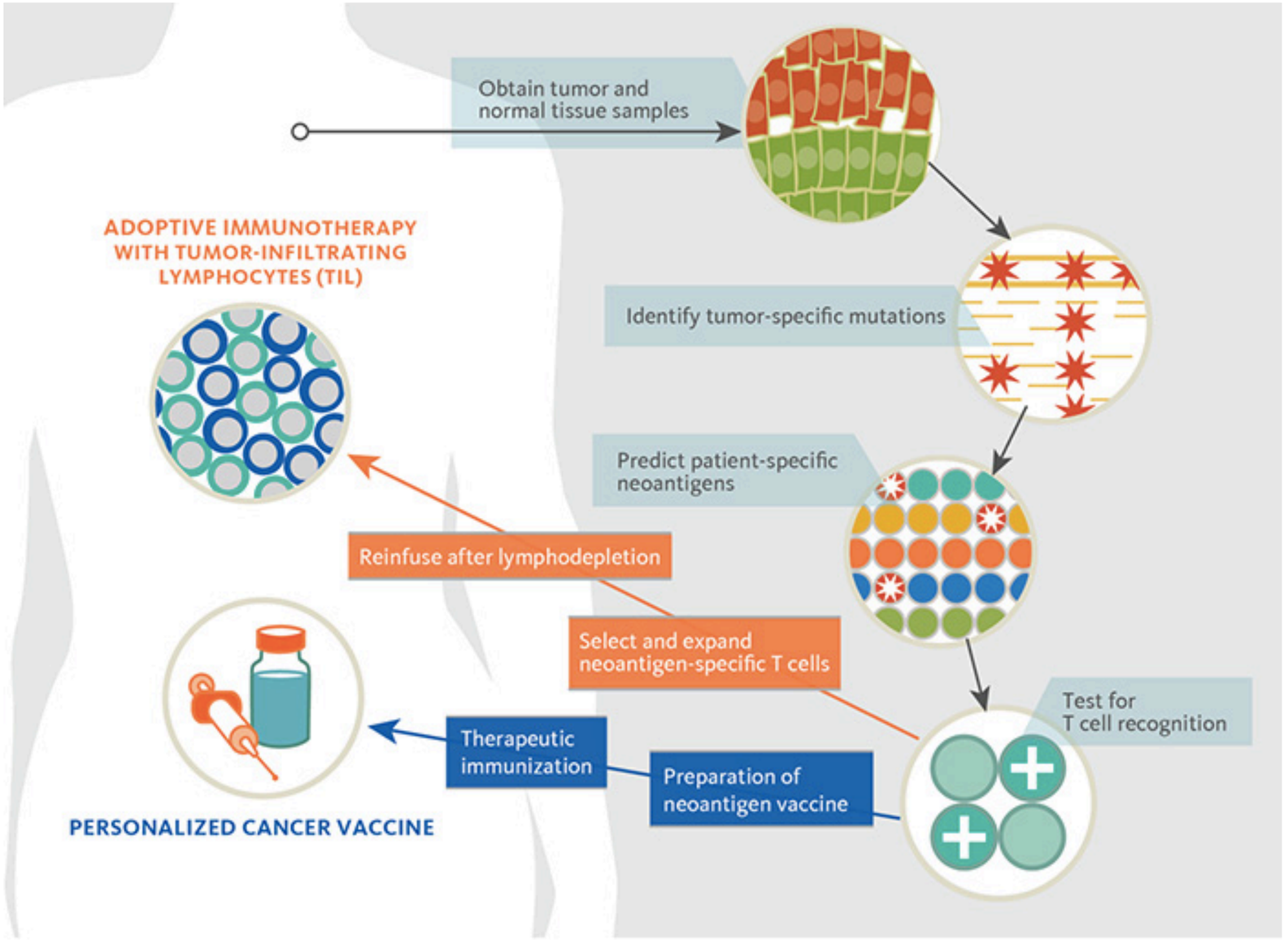
Andreas Wicki



Repeats in cancer and therapy

- DNA short Tandem Repeats (STRs) aka microsatellites

- not only for diagnostics but for immunotherapy!!!



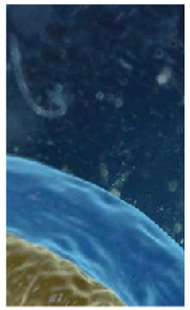
Immunotherapy Response for MSI-H Tumors

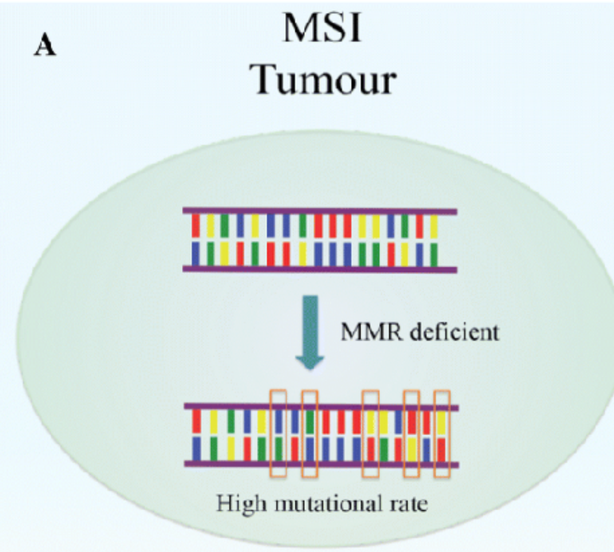
highly immunogenic because the rapid accumulation of mutations creates "neoantigens" (abnormal proteins) that trigger the body's immune system. This makes MSI-H tumors exceptionally responsive to immune checkpoint inhibitors.

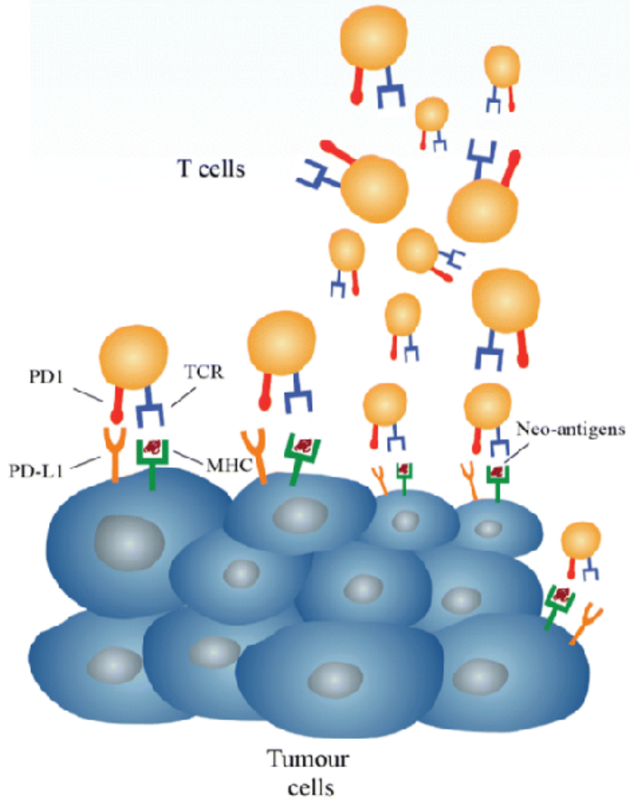
NIH NATIONAL CANCER INSTITUTE

Developing the First Precision Immunotherapy

In May 2017, the Food and Drug Administration (FDA) approved the first drug to treat tumors based on their genetic characteristics, regardless of where in the body the cancer originated. Until now, drugs have been approved based on their cell or tissue of origin, such as the breast or the lung. But pembrolizumab (Keytruda®) doesn't target the genetic abnormalities of cancer cells specifically—it targets the immune system.







Microsatellite instability in gastric cancer: molecular bases, clinical perspectives, and new treatment approaches

Some History in our lab

2020's:

Beyond Microsatellite Instability: Intrinsic Disorder as a Potential Link Between Protein Short Tandem Repeats and Cancer

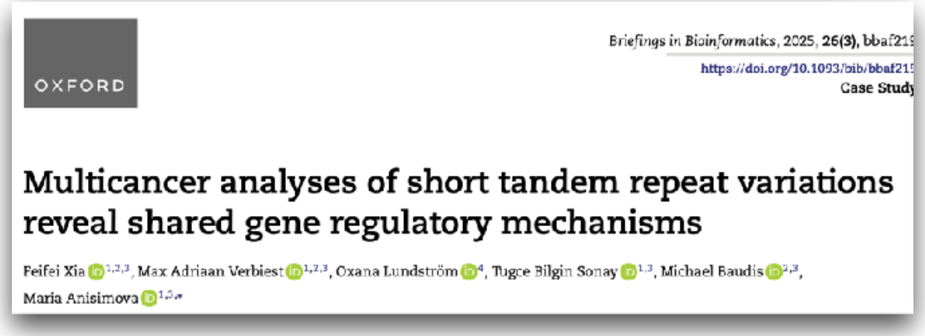
Max A. Verbiest[1,2,3], Matteo Delucchi[1,3], Tugce Bilgin Sonay[4] and Maria Anisimova[1,3]*


scientific reports

Check for updates

OPEN **Short tandem repeat mutations regulate gene expression in colorectal cancer**

Max A. Verbiest[1,2,3]✉, Oxana Lundström[1,4], Feifei Xia[1,2,3], Michael Baudis[2,3], Tugce Bilgin Sonay[2,3,5] & Maria Anisimova[1,3]✉


OXFORD

Briefings in Bioinformatics, 2025, 26(3), bbaf21

https://doi.org/10.1093/bib/bbaf21

Case Study

Multicancer analyses of short tandem repeat variations reveal shared gene regulatory mechanisms

Feifei Xia[1,2,3], Max Adriaan Verbiest[1,2,3], Oxana Lundström[4], Tugce Bilgin Sonay[1,3], Michael Baudis[2,3], Maria Anisimova[1,3,*]



2.3 M grant to work on STR patterns with deep learning


ORIGINAL RESEARCH article

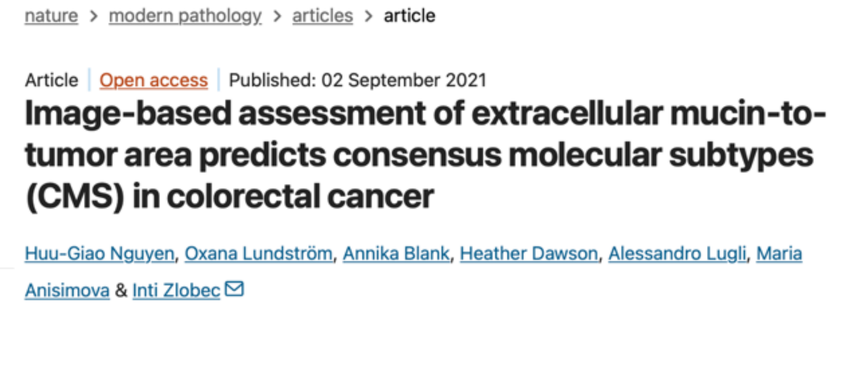
Front. Immunol., 08 January 2026

Sec. Cancer Immunity and Immunotherapy

Volume 16 - 2025 | https://doi.org/10.3389/fimmu.2025.1582464

Tumor mutational burden predicts neoantigen profiles and immunotherapy response in microsatellite stable tumors across different cancer types

Olesia Kondrateva[1,2,3*] Tugce Bilgin Sonay[1,2] Inti Zlobec[4] Maria Anisimova[1,2*]


nature > modern pathology > articles > article

Article | Open access | Published: 02 September 2021

Image-based assessment of extracellular mucin-to-tumor area predicts consensus molecular subtypes (CMS) in colorectal cancer

Huu-Giao Nguyen, Oxana Lundström, Annika Blank, Heather Dawson, Alessandro Lugli, Maria Anisimova & Inti Zlobec ✉

Method Development and Community Resources

Develop methods for STR analysis
We are developing an STR variant caller that incorporates structural variants to address open issues in STR genotyping

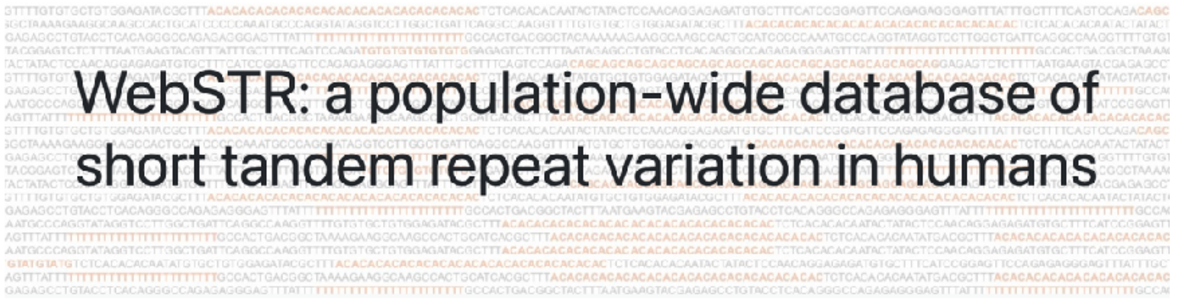
Contribute to community resources
The WebSTR database provides population-scale data of human STR variability

SIB Tandem Repeat Annotation Library

strvcf_annotator https://acg-team.github.io/strvcf_annotator/cli.html

ConSTRain: copy number guided STR allele inference

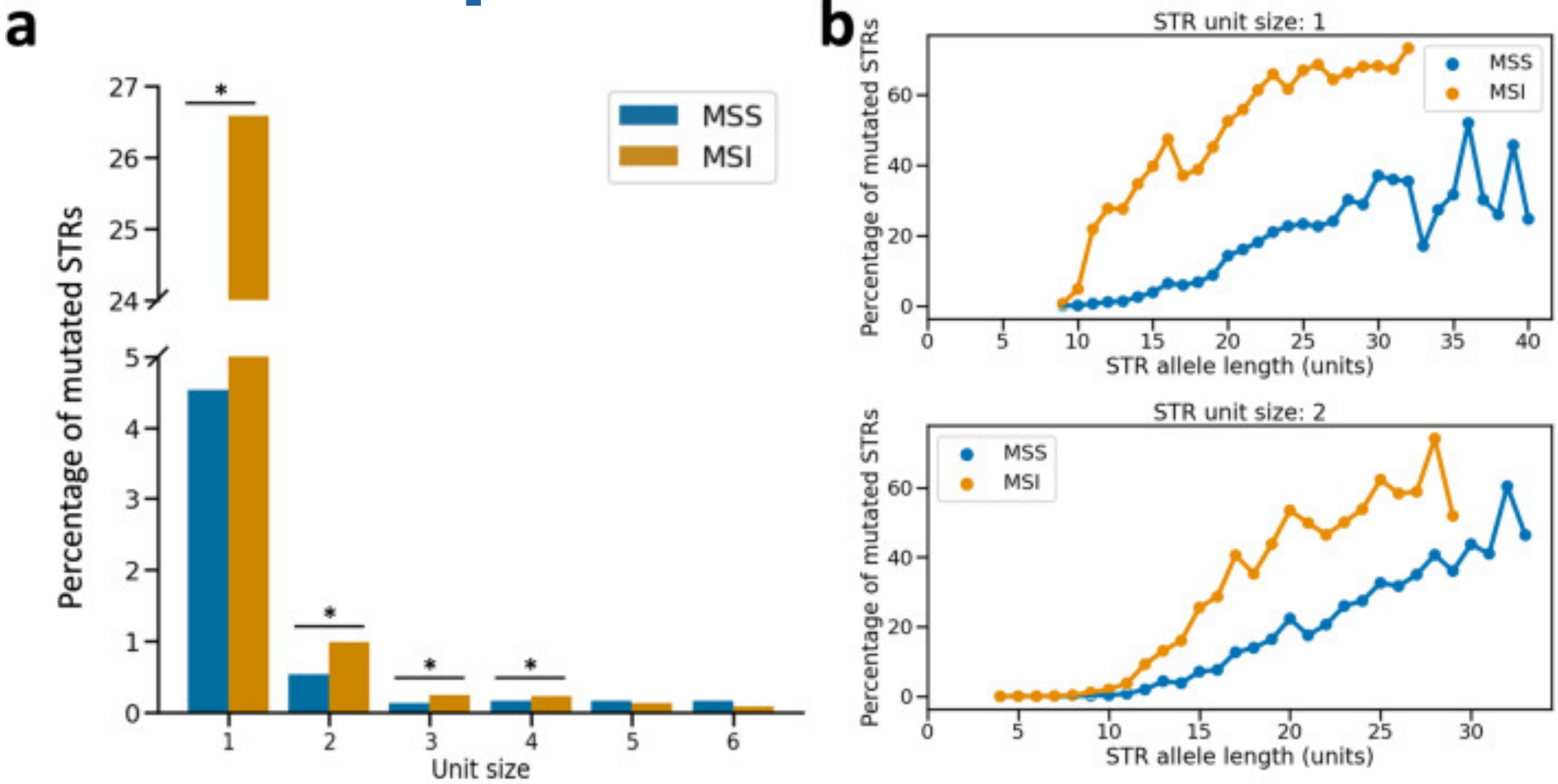
Extract STR signatures from annotated VCF https://acg-team.github.io/str_mut_signatures/



https://webstr.ucsd.edu/

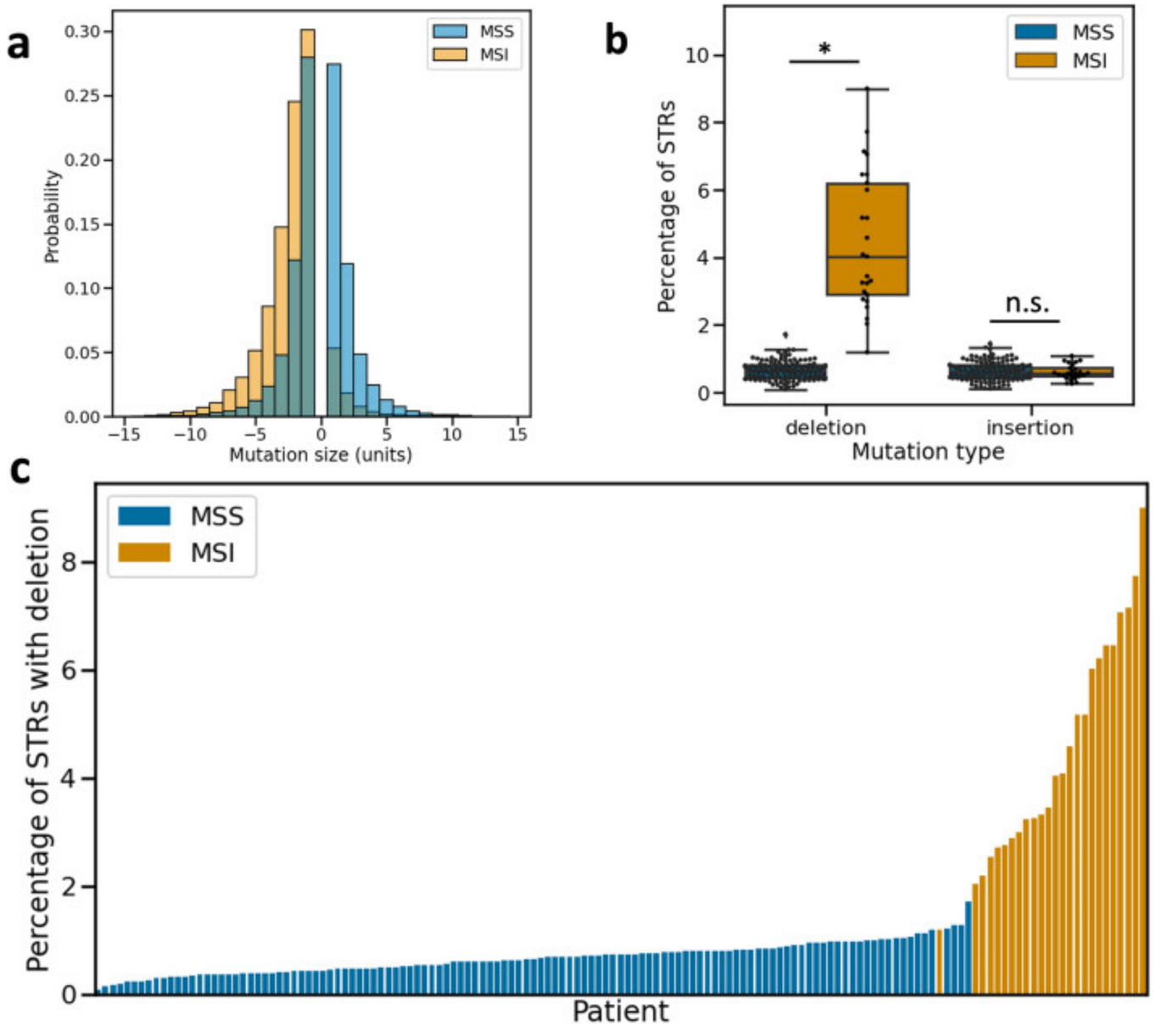
Delucchi et al. 2020

Mutated repeats in tumors



Verbiest et al. Nature Communications, 2024

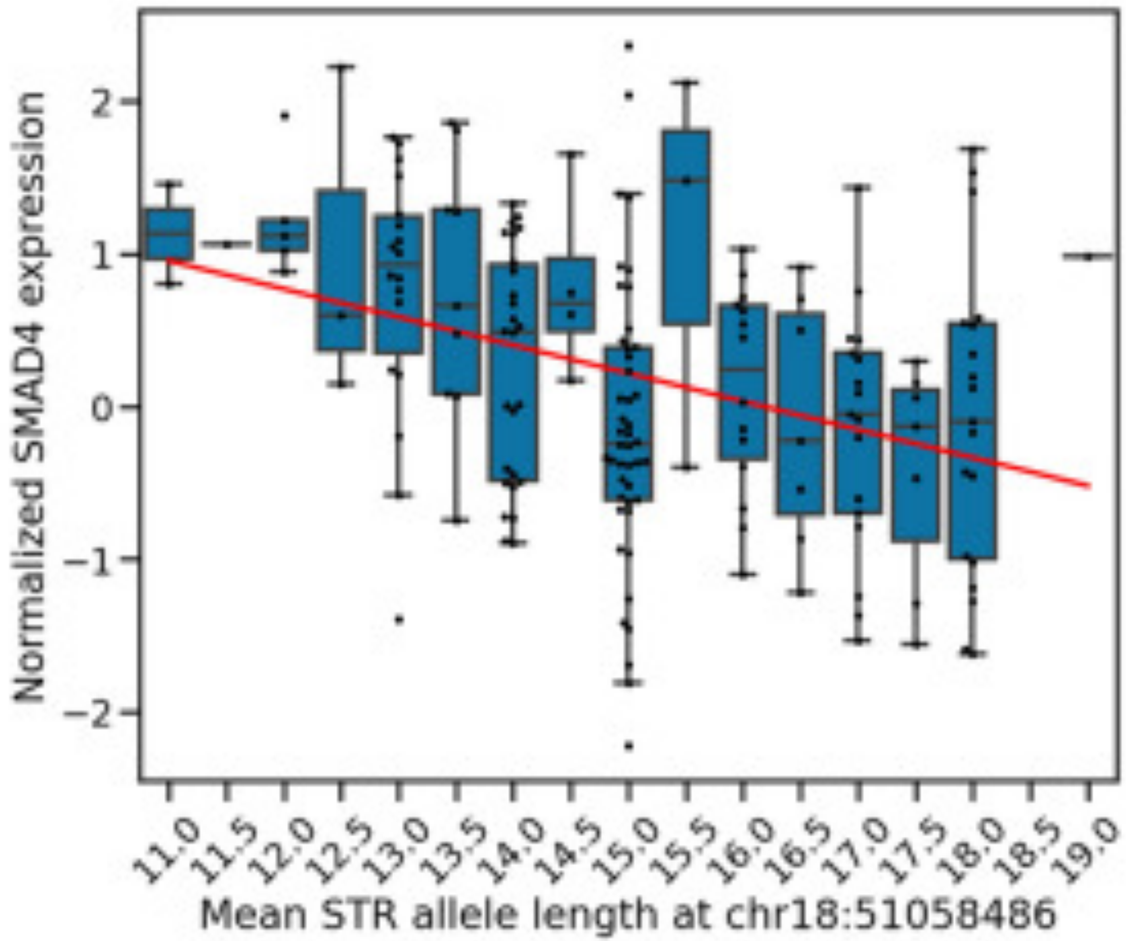
Lots of repeat deletion in tumors





Verbiest et al. Nature Communications, 2024

Identification of eSTRs in colorectal cancer



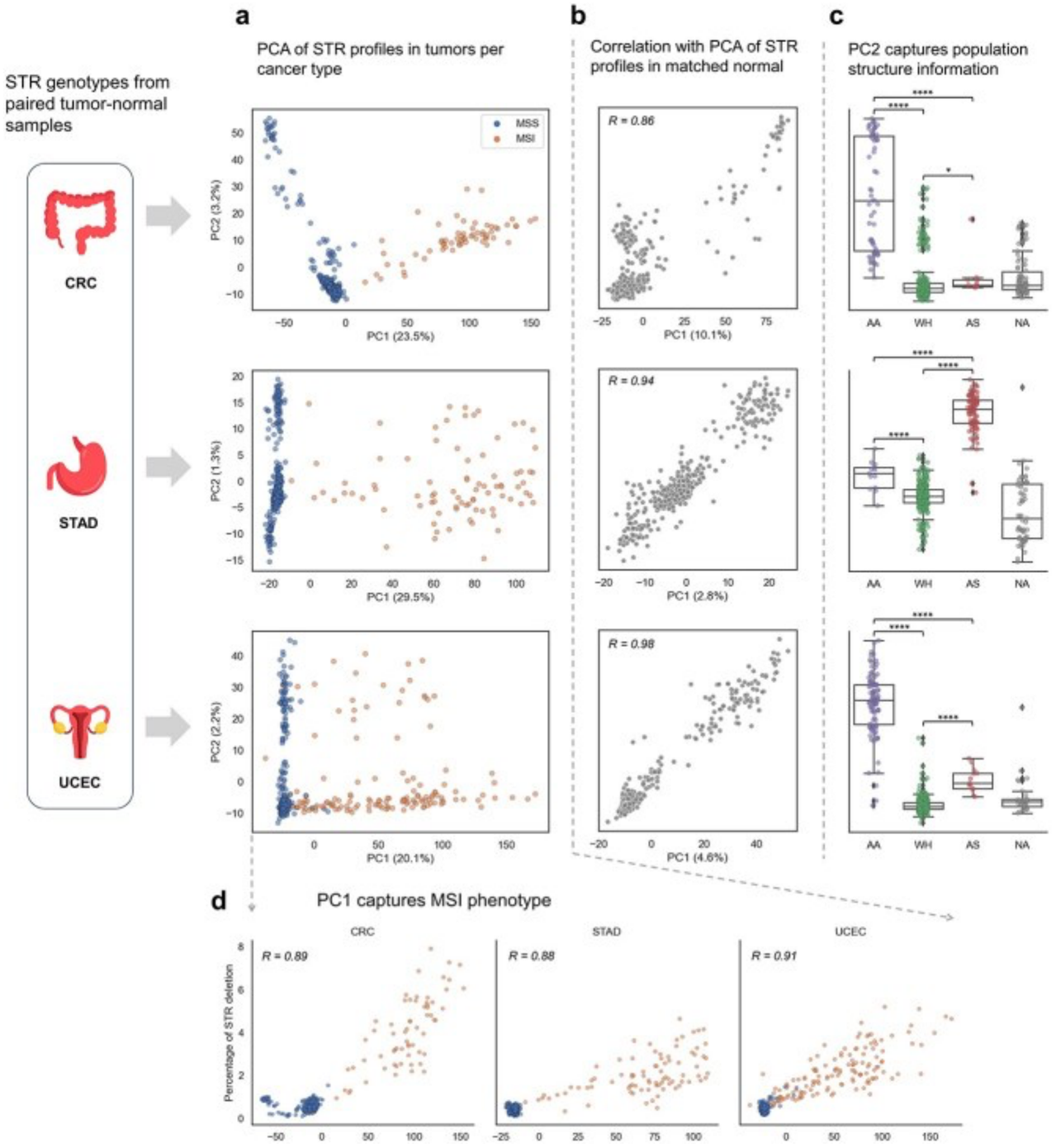

scientific reports

Check for updates

OPEN **Short tandem repeat mutations regulate gene expression in colorectal cancer**

Max A. Verbiest[1,2,3✉], Oxana Lundström[3,4], Feifei Xia[1,2,3], Michael Baudis[2,1], Tugce Bilgin Sonay[1,2,3] & Maria Anisimova[1,3✉]

Verbiest et al. Nature Communications, 2024



Extended the work to stomach and endometrial tumors

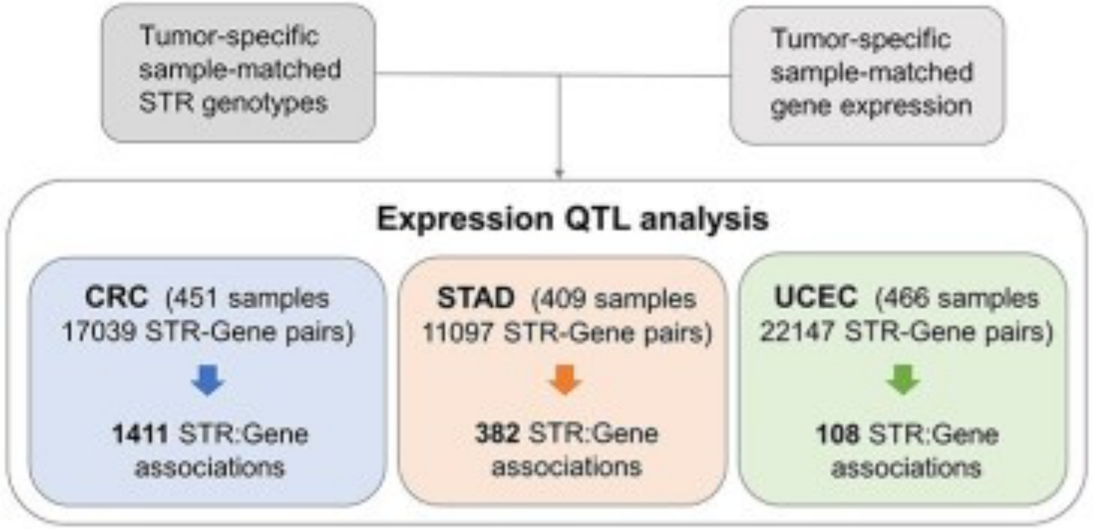
OXFORD

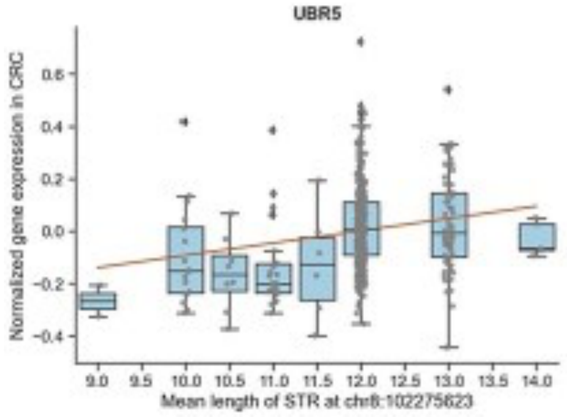
Briefings in Bioinformatics, 2025, 26(3), bbaf21
https://doi.org/10.1093/bib/bbaf21
Case Study

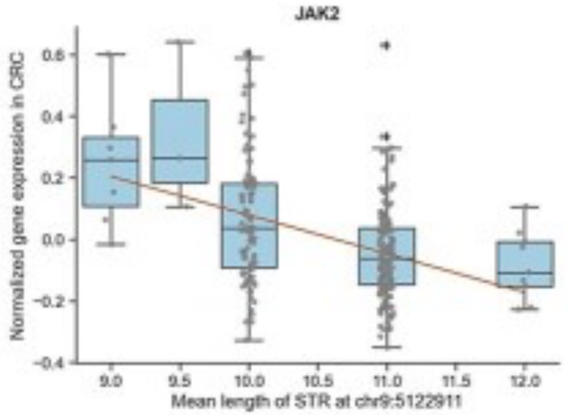
Multicancer analyses of short tandem repeat variations reveal shared gene regulatory mechanisms

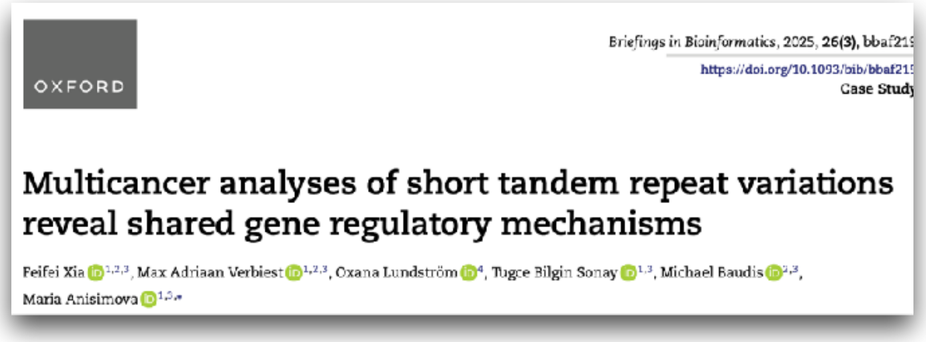
Feifei Xia, Max Adriaan Verbiest, Oxana Lundström, Tugce Bilgin Sonay, Michael Baudis, Maria Anisimova

Identification of eSTRs in stomach and endometrial cancer







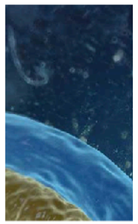


Meanwhile

NIH NATIONAL CANCER INSTITUTE

Developing the First Precision Immunotherapy

In May 2017, the Food and Drug Administration (FDA) approved the first drug to treat tumors based on their genetic characteristics, regardless of where in the body the cancer originated. Until now, drugs have been approved based on their cell or tissue of origin, such as the breast or the lung. But pembrolizumab (Keytruda®) doesn't target the genetic abnormalities of cancer cells specifically—it targets the immune system.



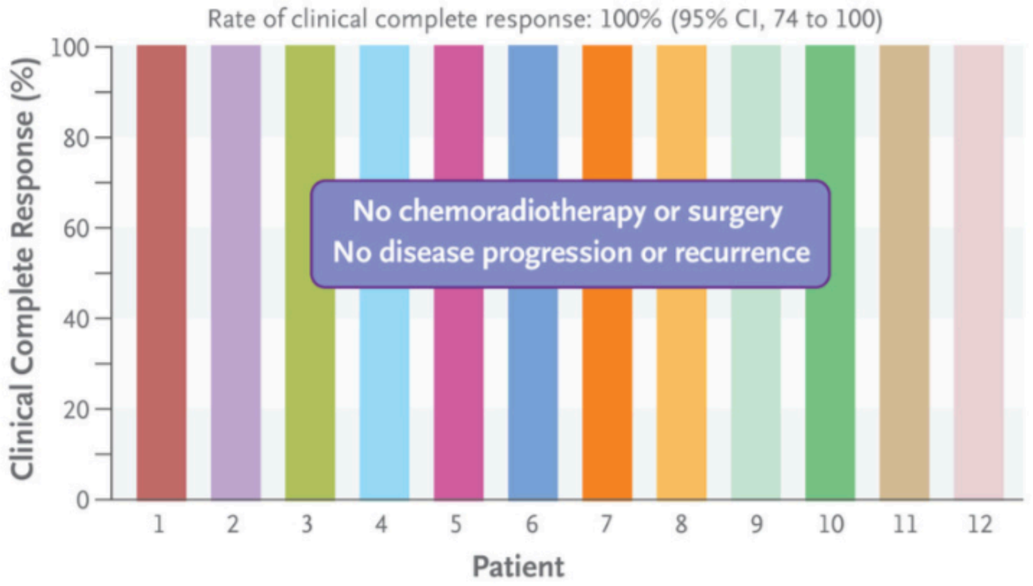
The NEW ENGLAND JOURNAL of MEDICINE

RESEARCH SUMMARY

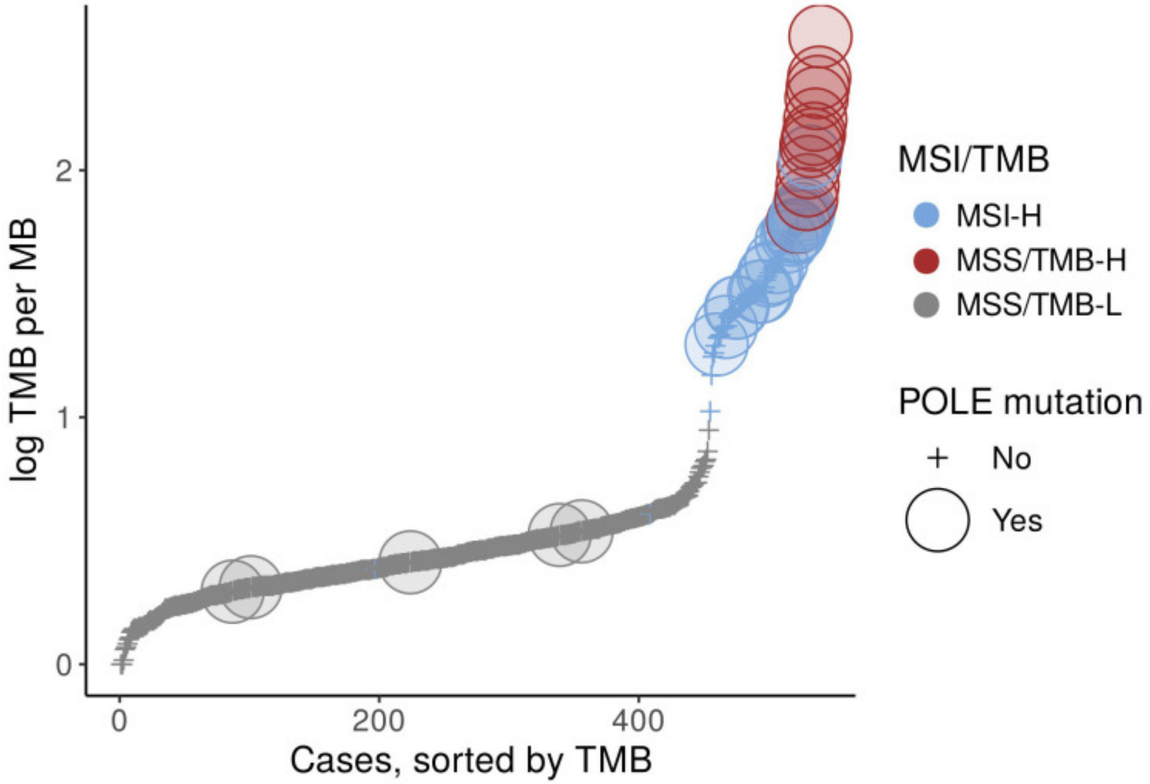
PD-1 Blockade in Mismatch Repair–Deficient, Locally Advanced Rectal Cancer

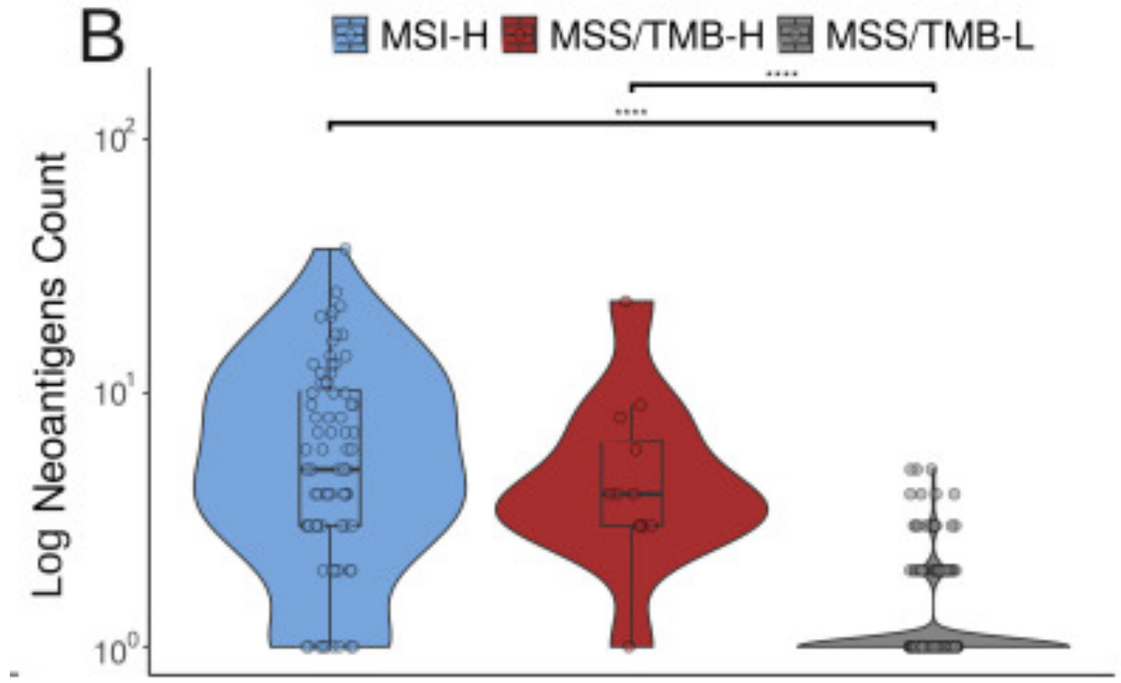
Cercek A et al. DOI: 10.1056/NEJMoa2201445

Overall Response to Dostarlimab in 12 Patients



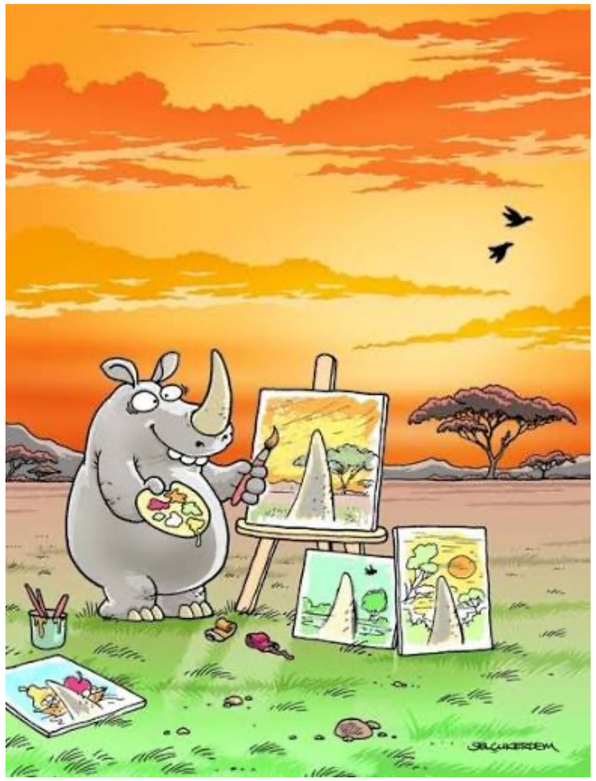
Identification of more subtypes to benefit from immunotherapy





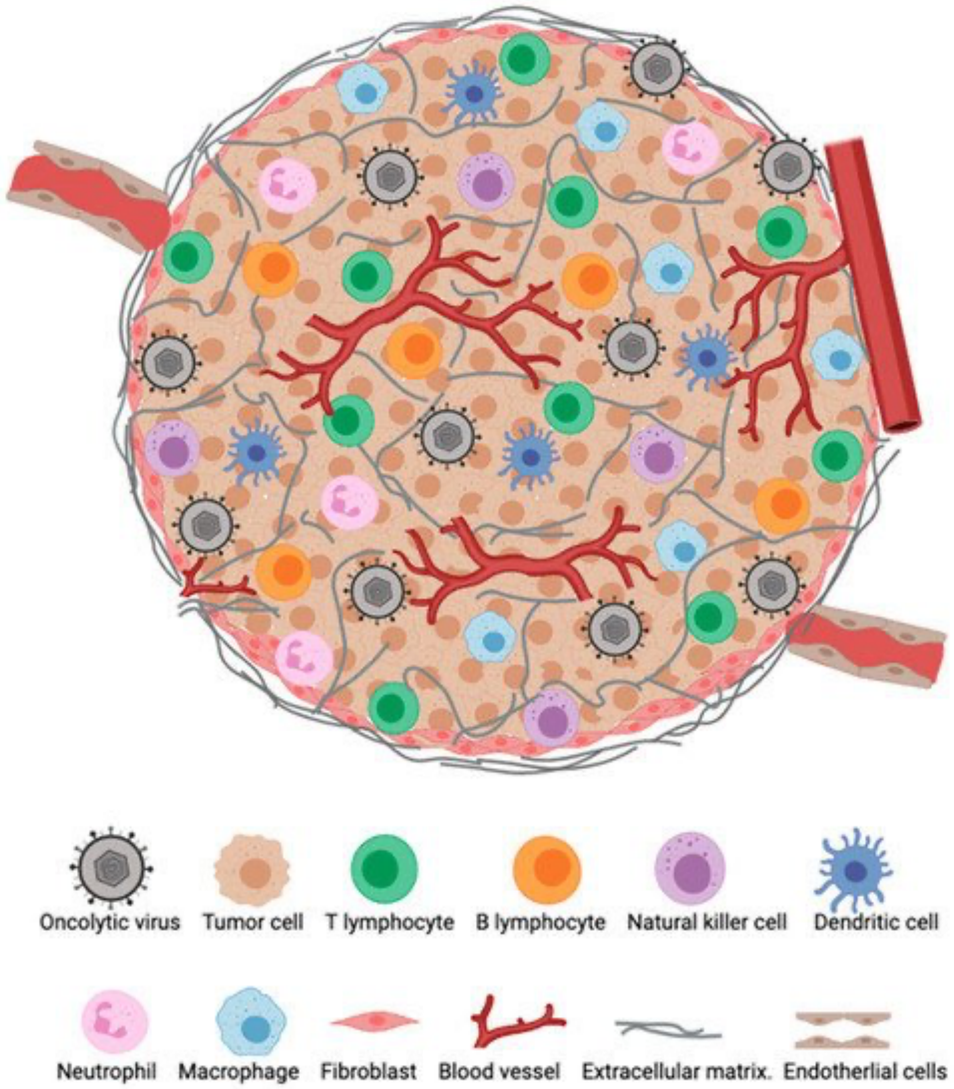
Kondrateva et al. Front Immunol, 2026

COLORECTAL CANCER BEYOND ITS DNA



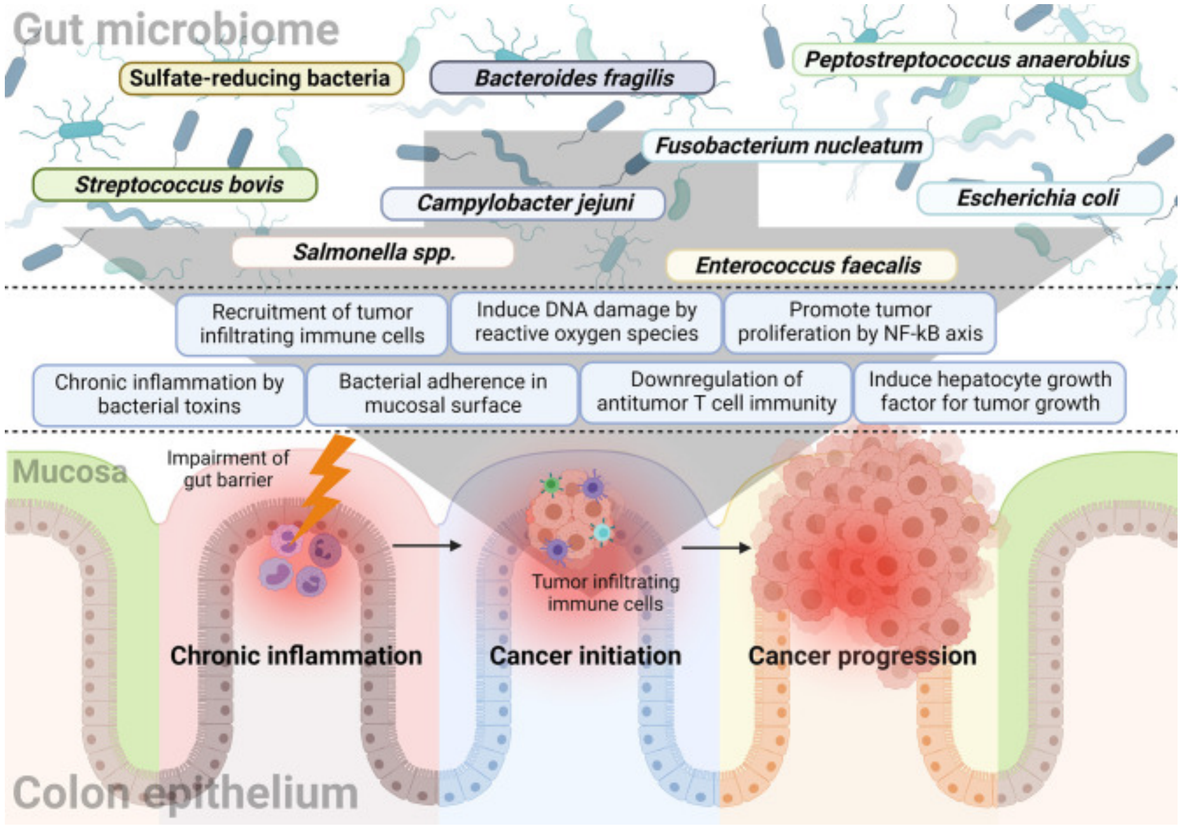
Tumor Microenvironmet

a complex ecosystem consisting of malignant cells, immune cells, cancer-associated fibroblasts, blood vessels, and the microbiome



Gut Microbiome and CRC

a complex ecosystem consisting of malignant cells, immune cells, cancer-associated fibroblasts, blood vessels, and the microbiome



Front Immunol. 2022 Jan 7;12:807648. doi: 10.3389/fimmu.2021.807648

Potential Role of the Gut Microbiome In Colorectal Cancer Progression

Jaeho Kim [1], Heung Kyu Lee [1,*]

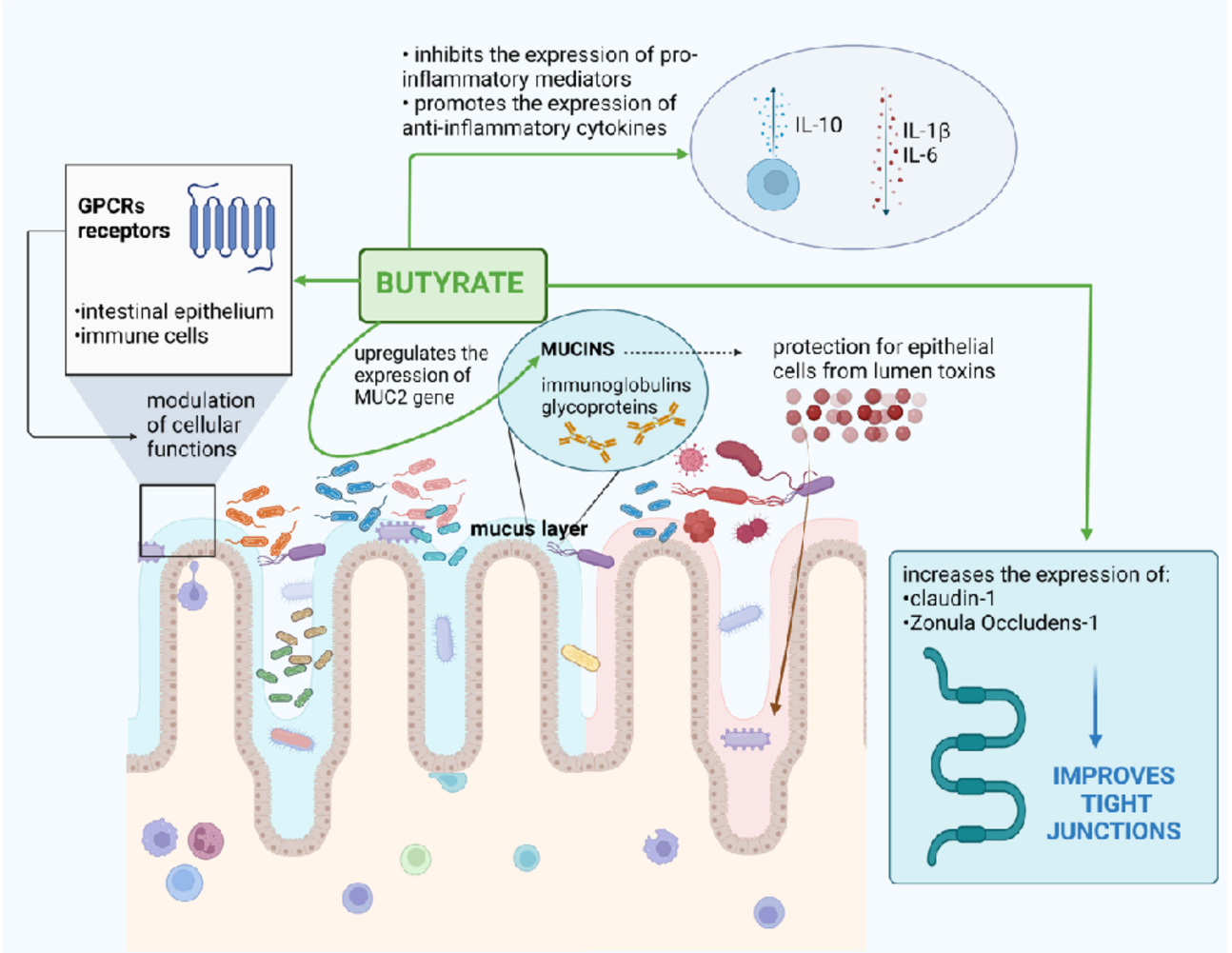
Specific harmful bacteria drive tumorigenesis by triggering chronic inflammation and inducing a tumor-favorable immune microenvironment.

Butyrate and Tumor Microenvironmet

Colorectal tumors are known to be depleted in their short-chain fatty acids. (SCFAs)

SCFAs especially butyrate play a critical role in preventing CRC.

They are produced by gut fermentation of dietary fiber. A diet high in insoluble fiber and resistant starches serves as the primary fuel for these microbes.



Sodium butyrate in both prevention and supportive treatment of colorectal cancer, 2022

CAN WE ~~TREAT~~ PREVENT COLORECTAL CANCER?

Further projects

- Machine Learning approaches to predict immunotherapy response based on tumor images and other omics
- population structure characterization based on STRs via genomic large language models
- method development to identify STR mutations in single cell omics data
- microbiome projects